National University of Computer and Emerging Sciences
Lahore Campus

Programming Fundamentals

Deadline: October 19, 2025

Course Instructor
Ms. Anosha Khan

Course Teacher Assistant
Muhammad Annus Shahzad

Assignment 3

Total Time (Days):	**7**
Total Marks:	**100**
Total Questions:	**6**

Instructions to be followed

Include comments in your code wherever necessary to improve clarity. The use of Artificial Intelligence tools or Large Language Models (LLMs), such as ChatGPT or DeepSeek, is strictly forbidden. Any student found using AI or LLMs to complete this assignment will receive **zero marks** and may face **serious disciplinary action (DC)** in the future.

If you are unsure about any part of the assignment, consult your **TA for clarification** before making assumptions.

Submission Instructions

Solve each question in a **separate C++ file**. Name each file using this format (RollNumber_Q1.cpp, RollNumber_Q2.cpp) Example: 25L_XXXX_Q1.cpp, 25_XXXX_Q2.cpp and so on. Put all your C++ files into **one ZIP file** with the following format **(25L_XXXX.zip)** and submit that **ZIP File** on Google Classroom.

Important: *Submissions that do not follow these instructions will result in* ***deduction of marks****. It is your duty to submit working code on GCR following the format given above. Students are expected to ensure that their programs* ***handle edge cases*** *with a user-friendly output if needed. Also handle* ***unwanted values & errors****.*

The syllabus includes Functions & Integer Arrays and topics covered before this. The use of advanced topics will lead to deduction of marks.

Your code must be highly ***modular*** *in nature. Code snippets that can be reused and seem complex to read & understand must be encapsulated in a function to increase readability and reusability. You are not allowed to create* ***global variables****. Each array shall be created in main function and passed via* ***parameter*** *to the function.*

Flip the Page for Questions

Question 01

Given an integer array, write a function to remove the **duplicates** in the array without making another array.

Question 02

Given an array of integers, write a function that returns the **k-th largest number** in the array.

Question 03

Given 2 sorted integer arrays. Create a function that combines these two sorted arrays into a single sorted array. *You are not allowed to use nested loops in this question.*

Question 04

Given a binary array[1], create a function that returns the maximum length of the sub array having consecutive 1's after deleting exactly 1 element. See the examples below

Examples

Input: [1,1,0,1] **Output:** 3 **Explanation** After deleting the number in position 2, [1,1,1] contains 3 numbers with value of 1's.	**Input:** [1,1,1] **Output:** 2 **Explanation** You must delete one element.	**Input:** [0,1,1,1,0,1,1,0,1] **Output:** 5 **Explanation** After deleting the number in position 4, [0,1,1,1,1,1,0,1] longest subarray with value of 1's is [1,1,1,1,1]

Question 05

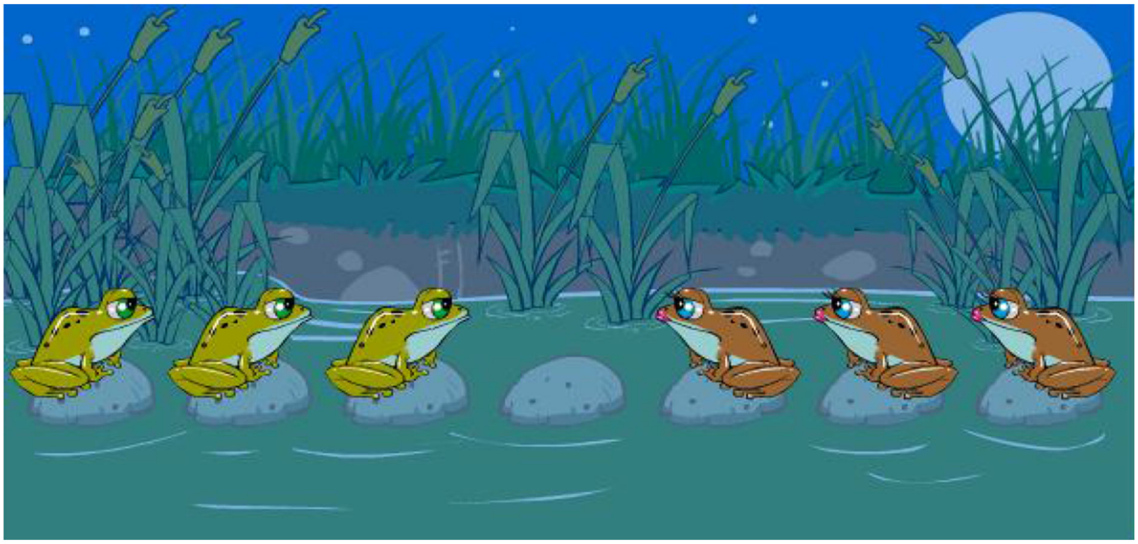
You are planning to start a new business of printing flexes. In the first step you have bought a printing machine. Every day you receive printing orders from **N** different customers. Customer $\mathbf{C_i}$ job requires time $\mathbf{t_i}$. The wait time of a customer $\mathbf{C_i}$ is $\mathbf{W_i}$ which is the amount of time the customer waited till its job completion. You have only one printing machine, so you have to devise a schedule of printing. You want your customers to be happy. The happiness of the customer is dependent on the waiting time. The smaller the wait time the happier your customer is. Devise a schedule that minimizes the total wait time of all the **N** customers.

Suppose there are three customers with job time $\mathbf{t_1}$ =4, $\mathbf{t_2}$ =5, $\mathbf{t_3}$ =2. If you schedule (1, 2, 3) then $\mathbf{w_1}$=4, $\mathbf{w_2}$=9, $\mathbf{w_3}$=11 total wait time is 4+9+11=24. There is a combination that would result in a total time less than this.

Create a function to **display the minimal schedule** and another function **to calculate the minimal waiting time** of all the **N** customers.

[1] A binary array is an array that stores only the values 1 and 0.

Question 06



There are seven stones. There are three frogs on three stones starting from the left and three frogs on the stones starting from right, one frog on each stone, so that the middle most stone is empty. Swap the frogs. 3 from the left have to jump on the 3 stones on the right and vice versa. Each frog can jump just on the adjacent stone or jump over another frog if there is an empty stone behind it. A frog can not jump over 2 frogs with no stone in between. Left frogs can only move Right. Right Frogs can only move left.

Your task is to perform jumps until all the right frogs are swapped with all the left frogs. Make sure to print each move as you make during the movement.

Represent left frogs with number 0 (L) and right frogs with number 1 (R).

The output should be as follows:

initial state: 0 0 0 _ 1 1 1

final state: 1 1 1 _ 0 0 0

(Hint: Use integer arrays for storing state, you can represent the empty stone with -1)

I hope you all will learn something new
while doing this Assignment
Best of Luck Guys
See you all energetic than before in Evaluation
Best Wishes